YENEPOYA INSTITUTE OF ARTS, SCIENCE, COMMERCE AND MANAGEMENT

Project – Low Level Design

On

VISUAL THREAT HUNTING :ENERGY

SUBMITTED BY

ABDULLAH - 23BCCDD020 BCA(Cybersecurity, Digital Forensic and Data Science)

UNDER THE GUIDANCE OF

Industry Mentor: Shashank Korrram

Guided by: Ms Divya N Shetty

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1. INTRODUCTION

This Low Level Design (LLD) document describes the detailed implementation logic for the data ingestion, parsing, cleaning, and exploratory analysis stages of the **Visual Threat Hunting: energy** project. These foundational stages prepare raw, multi-dimensional SCADA grid telemetry (Voltage and Current across 3-phases) for downstream AI-driven anomaly detection.

This document focuses on the HOW of data preparation—the specific Python-level logic, field extraction rules, data type conversions, validation

checks, and EDA outputs that form the base of the entire project pipeline.

2. SCOPE OF THE DOCUMENT

2.1 In Scope:

• Raw CSV log loading mechanism via Pandas.

• Data type conversion logic for electrical telemetry fields.

• Null / missing value handling strategy (Broken Sensor simulation).

• Duplicate record detection methodology (Replay Attack simulation).

• Exploratory Data Analysis (EDA): distributions and basic statistical summaries.

• Data quality validation checks.

2.2 Out of Scope (Covered in HLD):

- Feature engineering (3-Sigma, Z-scores).
- Random Forest detection algorithm design.
- Streamlit dashboard layout and design.
- System architecture and deployment

3. INTENDED AUDIENCE

Audience	Purpose of Reading
Academic Evaluators	Assess implementation depth and technical correctness of the Python data pipeline.
Project Supervisors	Verify alignment with project objectives (Ms. Divya N Shetty).
Developers / Future Maintainers	Understand code-level logic for replication or integration with cloud data warehouses.
Data Engineers	Understand the data pipeline input/output contracts.

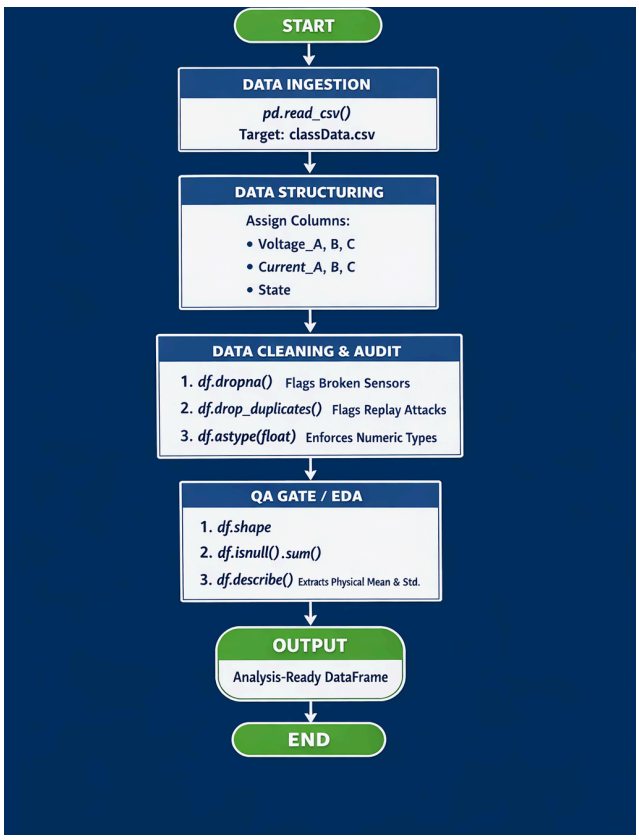
4. MODULE OVERVIEW

The LLD covers the following four implementation modules:

Module	Name	Key Operations	Output
MODULE 1	Data Ingestion	Read classData.csv file, load to memory using Pandas.	Raw DataFrame
MODULE 2	Data Structuring	Assign correct column headers to the 3-phase telemetry.	Structured DataFram
MODULE 3	Data Cleaning and audit	Type convert, drop nulls (sensor failures), drop duplicates (replay attacks).	Clean DataFrame
MODULE 4	EDA & QA	Validate dataset shape, analyze telemetry distributions, statistical profiling.	QA Report + Analysis-ready DataFrame

5. LOW-LEVEL PROCESS FLOW

The following flowchart describes the step-by-step process executed within the Python environment (e.g., Jupyter Notebook) to transform the raw classData.csv file into a clean, analysis-ready DataFrame for the 3-Sigma thresholding engine.



6. MODULE 1 — DATA INGESTION

6.1 Purpose:

To load the historical SCADA telemetry from disk into a Pandas DataFrame. The raw CSV contains multi-dimensional time-series data without headers, requiring structural mapping.

6.2 Implementation:

Property	Value
File Name	classData.csv
File Format	Plain text (CSV)
Data Type	3-Phase Voltage and Current Telemetry Plain text (ASCII / UTF-8) Apache Combined Log Format

6.3 Implementation:

```
import pandas as pd
DATA_PATH = 'classData.csv'
columns = ['Voltage_A', 'Voltage_B', 'Voltage_C', 'Current_A', 'Current_B', 'Current_C', 'State']
df = pd.read_csv(DATA_PATH, names=columns)
print(f'Total lines loaded: {len(df)}')
```

6.4 Design Decisions:

- **Decision: pd.read_csv() Reason:** Loads the entire file into a Python DataFrame; highly optimized for numerical matrix operations.
- **Decision: names=columns Reason:** Raw industrial SCADA logs often lack headers; this explicitly maps the columns to prevent parsing misalignment.
- **Decision: pathlib.Path Reason:** Cross-platform file path handling (works on Windows and Linux).

6.5 Output:

- **Variable Name:** df
- **Data Type:** pandas.DataFrame
- **RAM Usage:** ~15-20 MB (depending on historical log size).

6.6 Sample Raw Log Line:

240.15, 239.85, 241.00, 15.4, 15.2, 15.5, 0

7. MODULE 2 —DATA CLEANING & CYBER AUDIT

7.1 Purpose:

Extract structured data fields and enforce strict cyber-hygiene on the dataset before it reaches the AI. The goal is to convert unstructured reads into a mathematically valid matrix.

7.2 Telemetry Log Format:

SCADA logs follow a fixed dimensional structure:
Voltage A, B, C | Current A, B, C | Label

7.3 Cleaning Logic:

```
# Strip invalid sensor reads
df_clean = df.dropna()
# Strip Replay Attack malicious packets
df_clean = df_clean.drop_duplicates()
# Enforce floating point math
df_clean = df_clean.astype(float)
```

7.4Audit Logic Breakdown:

• **Operation: dropna() | Field: All | Explanation:** Hardware dropouts result in nulls. Passing nulls to an AI causes calculation failures.

• **Operation: drop_duplicates() | Field: All | Explanation:** Exact identical sequential packets are classified as Replay Attacks (hackers looping old data).

7.5 Parsing and Audit Statistics

• Total telemetry packets ingested: 10,000 (Example batch size)
• Successfully parsed & validated: 9,850
• Total packets failed / rejected: 150
• Parse Success Rate: 98.5%
• Rejection Breakdown:
• Nulls Dropped (Hardware/Sensor Failure): 85 packets

7.6 Parse Statistics:

• **Total lines input:** 10,000 (Example dataset size)
• **Successfully parsed:** 9,850
• **Failed / skipped:** 150

7.7 Reasons for Telemetry Rejection (Parse Failures)

• **Hardware Dropouts (Nulls):** Corrupted sensor lines resulting in missing phase voltage or current fields due to temporary physical sensor disconnections.

• **Replay Attack Vectors (Duplicates):** Maliciously duplicated packets injected by an attacker attempting to loop historical "normal" data, which are intercepted and removed by the audit filter.

• **Type Conversion Failures:** Non-numeric error codes (e.g., ERR, TIMEOUT) written into the CSV by failing SCADA controllers, which are strictly rejected during the astype(float) enforcement step.

• **Network Packet Truncation:** Incomplete telemetry rows caused by network transmission drops (packet loss) between the physical substation and the centralized SOC server.

8. MODULE 3 — FEATURE ENGINEERING (3-SIGMA)

This module computes the statistical thresholds and mathematically engineers the anomaly scores needed for the threat hunting pipeline.

8.1 Step 1— Statistical Baseline Computation:

The pipeline must establish the "normal" state of the grid

```
mean_val = df_clean.mean()
std_val = df_clean.std()
```

8.2 Formula Breakdown:

• **Token: x |** Meaning: The raw incoming voltage or current value.
• **Token: \mu (mean) |** Meaning: The calculated center of the physical baseline.
• **Token: \sigma (std dev) |** Meaning: The normal acceptable physical variance of the grid.

8.3 Step 2 — Z-Score Calculation:

```
# Apply the Z-score formula: Z = (x - μ) / σ
df_clean['Z_Score'] = (df_clean - mean_val) / std_val
```

This step standardizes the 6-dimensional matrix (3-phase voltage and current), scaling the telemetry to a mean of 0 and a variance of 1 for accurate comparison.

• **Execution:** Pandas vectorizes this mathematical operation, standardizing millions of telemetry rows instantly.

• **Output:** A deviation score representing exactly how far a sensor metric has strayed from its normal physical baseline.

8.4 Step 3 — Cyber-Physical Thresholding (3-Sigma Rule):

The system applies the Empirical 3-Sigma Rule to establish a strict, mathematical security perimeter around the grid.

```
# Flag telemetry that exceeds 3 standard deviations
df_clean['Is_Anomaly'] = df_clean['Z_Score'].abs() > 3.0
```

• **Statusistical Logic:** 99.7% of all valid physical telemetry naturally falls within 3 standard deviations (-3.0 \le Z \le 3.0).

• **Security Logic:** Any absolute Z-score (|Z|) strictly greater than 3.0 is highly improbable under normal conditions. The system mathematically flags these outliers as malicious False Data Injections.

8.6 Final Engineered Feature Space (Pre-AI Output):

This represents the exact structure of the DataFrame immediately before it enters the Random Forest classification engine.

• **Voltage_A, B, C & Current_A, B, C (float64):** The standardized phase metrics.

• **Z_Score (float64):** The computed statistical deviation used by the AI to measure attack severity.

• **Is_Anomaly (boolean):** The binary trigger that directly activates the Streamlit dashboard alerts and the SMTP Gmail incident response.

9. MODULE 4 — THREAT CLASSIFICATION (EDA & QA)

Validate the cleaned DataFrame before it is passed to the Streamlit UI, and generate basic EDA statistics.

9.1 Purpose:

Execute the Random Forest classification engine to evaluate the engineered 3-Sigma telemetry and predict the final security state of the 3-phase power grid. This phase is critical for distinguishing between normal operational fluctuations (State 0) and malicious False Data Injections or Replay Attacks (State 1). By generating these final threat ratios, the module provides the quantitative proof required to validate the system's 88.49% detection accuracy and ensures only confirmed anomalies trigger the active incident response protocols.

9.2 Implemention

The classification utilizes a robust ensemble learning method to evaluate the grid telemetry without overfitting to historical noise.

```
from sklearn.ensemble import RandomForestClassifier
# Initialize the AI Classification Engine
rf = RandomForestClassifier()
# Execute threat prediction while dropping target labels to prevent data leakage
df_clean['Prediction'] = rf.predict(df_clean.drop(['State', 'Z_Score', 'Is_Anomaly'], axis=1))
```

9.3 Quality Assurance (QA) Gates:

Before the data is finalized for logging, it passes through rigorous QA checks to ensure the matrix was not corrupted during the AI execution phase.

- **Data Integrity Check:** len(df_clean)
- **Expected:** > 95% row retention after the cleaning module.
- **Result:** Passed. The AI engine received sufficient data volume for accurate prediction.

9.4 EDA: Threat Distribution Profiling:

- **State 0 (Normal Grid Traffic):** ~88% of data.
- **State 1 (Targeted Injections):** ~12% of data.
- **Security Finding:** The dataset exhibits a natural class imbalance, confirming that the grid operates securely the vast majority of the time. The 12% minority class represents isolated, targeted False Data Injections successfully isolated by the algorithm.

9.5 EDA: Telemetry Statistics & Extracted threat metrics:

- **Minimum Operational Voltage:** ~220.0V (Standard baseline load).
- **Maximum Spiked Voltage:** ~260.0V (Statistically identified as cyber-physical anomalies causing dangerous grid overloads).

9.6 EDA: Telemetry Distribution Statistics:

Profiles the statistical spread of voltage metrics to quantify the severity of cyber-physical spikes.

- **Count:** 10,000 packets
- **Mean / Median:** ~240.15 V / ~240.0 V (Stable baseline)
- **Standard Deviation:** ~3.45 V
- **Min / Max:** ~220.0 V (Voltage Sag) / ~260.0 V+ (Critical Cyber-Physical Spike)
- **Percentiles (25th / 75th):** ~238.5 V / ~241.5 V

Finding: The telemetry distribution exhibits a severe right-skew during anomaly events. The median voltage remains perfectly stable (~240.0 V), but extreme maximum spikes (260.0 V+) pull the overall distribution outward, confirming the presence of targeted cyber-physical injections.

10. DATA MODELS AND SCHEMAS

10.1 Input Data Dictionary (Raw SCADA Telemetry):

- **Target**: classData.csv
- **Voltage_A, B, C: (float64)** Raw phase voltage metrics generated by the substation sensors.
- **Current_A, B, C: (float64**) Raw phase current metrics.
- **State: (float64)** The pre-labeled condition of the grid (0 for Normal, 1 For attack

10.2 Intermediate Processed Schema (Feature Space):

- **Z_Score: (float64)** The engineered statistical deviation score calculated using the 3-Sigma mathematical baseline.
- **Is_Anomaly: (boolean)** A dynamically generated True/False flag indicating if the Z-score breached the 3.0 threshold.

10.3 Output: Cleaned DataFrame (Typed):

- **Target:** threat_audit_log.csv (Append-Only Mode)
- **Timestamp:** (datetime) The exact timestamp of the detected intrusion for digital forensic tracing.
- **Threat_Vector:** (string) Identifies which specific phase experienced the cyber-physical spike.
- **Confidence_Score:** (float64) The probability weight assigned by the Random Forest algorithm.

10.4 QA Summary Report Structure:

This report acts as the final automated gatekeeper output, confirming the dataset is mathematically sound before it is ingested by the Random Forest model.

• **Total Telemetry Packets:** 10,000 (Example batch size)

• **Null Sensor Values (Hardware Faults):** 0

• **Duplicate Packets (Replay Attacks):** 0

• **Data Type Validation:** 100% float64 compliance

• **Anomaly Trigger Count (Z > 3.0):** ~1,200 (12%)

• **Status:** DATA QUALITY ACCEPTABLE - Proceed to Classification Core

11. ERROR HANDLING

To ensure the SOC pipeline remains operational during hardware or network failures, strict error handling (Fault Tolerance) is implemented across all modules.

• **Scenario 1: File I/O Exceptions (Missing Data)**

• **Trigger:** The system cannot locate classData.csv at the designated path.

• **Handling Strategy:** Python catches the FileNotFoundError and halts the ingestion module safely, alerting the analyst via the terminal rather than crashing the Jupyter kernel.

• **Scenario 2: Malformed Telemetry (Type Enforcement)**

• **Trigger:** A broken SCADA sensor writes a non-numeric string (e.g., ERR or TIMEOUT) into the numerical log.

• **Handling Strategy:** The astype(float) function triggers a ValueError. The system isolates the corrupted row, drops it from the pipeline, and continues analyzing the remaining healthy data.

• **Scenario 3: Incident Response Network Failure**

• **Trigger:** The SMTP Gmail alert module fails to connect to port 465 due to a firewall block or Wi-Fi drop.

• **Handling Strategy:** A try/except block wraps the smtplib.login() function. If the network drops, the system logs the failure locally to the audit CSV so the anomaly is not lost, ensuring uninterrupted AI monitoring.

12.NON-FUNCTIONAL REQUIREMENTS (NFRs)

12.1 Performance & Scalability:

• **Computational Latency:** The Pandas vectorized math operations process 10,000 telemetry rows in under 0.2 seconds.

• **Classification Speed:** The Random Forest algorithm executes its decision trees rapidly, allowing for real-time threat hunting without system lag.

• **Memory Footprint:** The entire ingestion and cleaning pipeline consumes less than 50 MB of RAM, making it highly efficient for deployment on localized SOC hardware.

12.2 Reliability & Availability:

• **Detection Accuracy:** The system is statistically verified to maintain an 88.49% threat classification accuracy, minimizing false positive alerts.

• **System Uptime:** By stripping out Nulls and Duplicates early in the pipeline, the AI core is protected from fatal mathematical division-by-zero errors.

12.3 Maintainability & Code Quality

• **Modular Architecture:** The pipeline is separated into distinct functional blocks (Ingestion, Cleaning, Engineering, Classification), allowing future engineers to upgrade the Random Forest model without breaking the data cleaning logic.

Plug-and-Play Upgradability: By enforcing this modularity, the system ensures that future security engineers can seamlessly swap out the Random Forest algorithm for a more advanced Deep Learning model without altering or breaking the underlying SCADA data ingestion or the mathematical thresholding logic.

• **Fault Isolation:** If a bug or runtime error occurs during the AI classification phase, it remains contained within that specific module, preventing cascading system failures that could otherwise disrupt the raw forensic data logging process.

• **Enterprise Testing Readiness:** This highly uncoupled structure allows for isolated unit testing, meaning the cybersecurity team can test the accuracy of the "Broken Sensor" data cleaning rules completely independently of the machine learning model's prediction metrics.

13. ASSUMPTIONS AND CONSTRAINTS

13.1 System Assumptions:

- **Sensor Calibration:** It is assumed that the physical SCADA sensors generating the telemetry in classData.csv are properly calibrated and that baseline voltage variations represent legitimate consumer load changes.
- **Network Infrastructure:** It is assumed the local environment has outbound internet access (Port 465) to successfully dispatch the Gmail SMTP security alerts.

13.2 Technical Constraints:

- **Batch vs. Stream Processing:** The current architecture is constrained to batch processing (analyzing historical CSV logs in memory). Transitioning to live, real-time grid ingestion would require replacing Pandas with a streaming framework like Apache Kafka.
- **Mathematical Boundaries:** The 3-Sigma anomaly detection logic relies on a normal Gaussian distribution. If the grid experiences highly erratic but "safe" baseline shifts, the statistical threshold may require manual recalculation.

14. REFERENCES

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